WORKED SOLUTION

# STEP III 2009, Question 3

Pure • Limits and Curve Sketching

**STEP III 2009, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q3

## The key idea

The function is undefined at $t = 0$ only because of a $0/0$, so power series tell us what it "should" be there. Part (ii) is a matter of putting everything over a common denominator; the even function that appears is $\frac{t}{2}\coth\frac{t}{2}$ in disguise. For part (iii), the sign of $f'(t)$ is decided entirely by its numerator, and that numerator is exactly $\mathrm{e}^t(1-t)-1$, which is why the inequality is there.

## Solution

### Part (i)

Using $\mathrm{e}^t = 1 + t + \frac{1}{2}t^2 + \frac{1}{6}t^3 + \cdots$, for $t \neq 0$ we may cancel a factor of $t$:

$$f(t) = \frac{t}{t + \frac{1}{2}t^2 + \frac{1}{6}t^3 + \cdots} = \frac{1}{1 + \frac{1}{2}t + \frac{1}{6}t^2 + \cdots} \longrightarrow 1 \quad \text{as } t \to 0,$$

since the denominator tends to 1. So $\lim_{t\to 0} f(t) = 1$.

By the quotient rule,

$$f'(t) = \frac{(\mathrm{e}^t - 1) - t\,\mathrm{e}^t}{(\mathrm{e}^t - 1)^2}.$$

Both numerator and denominator tend to 0, so expand each. The numerator is

$$\begin{aligned}\mathrm{e}^t - 1 - t\,\mathrm{e}^t &= (t + \tfrac{1}{2}t^2 + \tfrac{1}{6}t^3 + \cdots) - (t + t^2 + \tfrac{1}{2}t^3 + \cdots) \\ &= -\tfrac{1}{2}t^2 - \tfrac{1}{3}t^3 + \cdots,\end{aligned}$$

and the denominator is $(t + \frac{1}{2}t^2 + \cdots)^2 = t^2(1 + \frac{1}{2}t + \cdots)^2$. Dividing top and bottom by $t^2$,

$$f'(t) = \frac{-\frac{1}{2} - \frac{1}{3}t + \cdots}{(1 + \frac{1}{2}t + \cdots)^2} \longrightarrow -\frac{1}{2} \quad \text{as } t \to 0.$$

**Part (ii)**

Let $g(t) = f(t) + \frac{1}{2}t$. Over a common denominator,

$$g(t) = \frac{t}{e^t - 1} + \frac{t}{2} = \frac{2t + t(e^t - 1)}{2(e^t - 1)} = \frac{t\,(e^t + 1)}{2(e^t - 1)}.$$

Now replace $t$ by $-t$ and multiply top and bottom by $e^t$:

$$g(-t) = \frac{-t\,(e^{-t} + 1)}{2(e^{-t} - 1)} = \frac{-t\,(1 + e^t)}{2(1 - e^t)} = \frac{t\,(e^t + 1)}{2(e^t - 1)} = g(t).$$

So $f(t) + \frac{1}{2}t$ is even, as required. (Dividing top and bottom by $e^{t/2}$ shows $g(t) = \frac{t}{2}\coth\frac{t}{2}$, a product of two odd functions, which is another way to see it.)

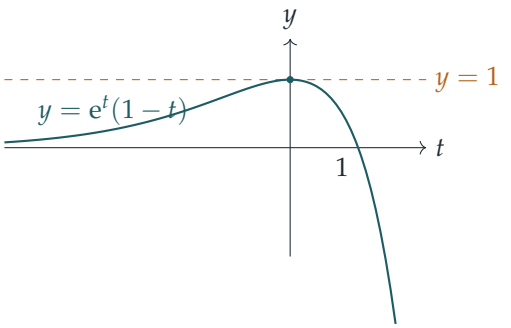
A useful consequence for the sketch: $f(t) - f(-t) = -t$, so the graph of $f$ is the graph of an even function tilted by the line $y = -\frac{1}{2}t$.

**Part (iii)**

Let $h(t) = e^t(1 - t)$. Then

$$h'(t) = e^t(1 - t) - e^t = -t\,e^t,$$

which is positive for $t < 0$ and negative for $t > 0$. So $h$ increases up to $t = 0$ and decreases afterwards: its only turning point is a maximum, $h(0) = 1$. Also $h(t) \to 0$ as $t \to -\infty$, and $h(t) < 0$ for $t > 1$.




The curve touches $y = 1$ at $t = 0$ and lies strictly below it everywhere else, so

$$e^t(1 - t) \le 1, \quad \text{with equality only at } t = 0.$$

**Ask yourself** How does this inequality tell us anything about $f'(t)$?

Look again at the numerator of $f'(t)$ from part (i):

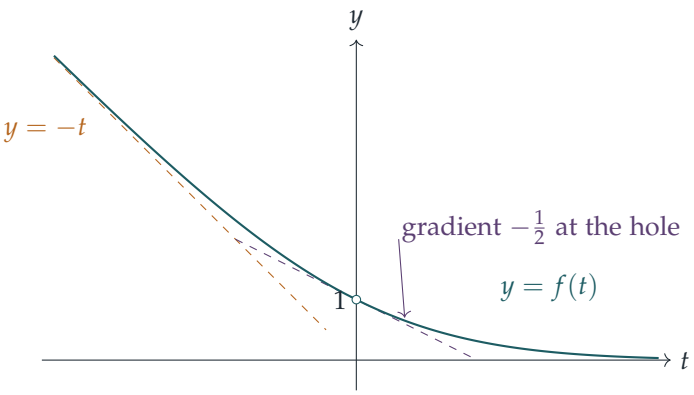
$$e^t - 1 - t\,e^t = e^t(1 - t) - 1.$$

For $t \neq 0$ this is strictly negative, by the strict inequality above. The denominator $(e^t - 1)^2$ is strictly positive for $t \neq 0$. Hence $f'(t) < 0$ for every $t \neq 0$; in particular $f'(t) \neq 0$, so $f$ has no stationary points and is decreasing on each side of 0.

**The sketch of $f$**

Collect the facts.

- $f(t) > 0$ for all $t \neq 0$, since $t$ and $\mathrm{e}^t - 1$ have the same sign.
- Near 0 the graph approaches $(0, 1)$ with gradient $-\frac{1}{2}$ (part (i)); the point itself is missing, as $f(0)$ is undefined.
- $f$ is decreasing throughout (part (iii)).
- As $t \to \infty$, $f(t) = t/(\mathrm{e}^t - 1) \to 0^+$, because the exponential beats $t$.
- As $t \to -\infty$, $f(t) + t = \dfrac{t\,\mathrm{e}^t}{\mathrm{e}^t - 1} \to 0$, so $y = -t$ is an oblique asymptote. This also follows from part (ii): $f(t) = f(-t) - t$ and $f(-t) \to 0$.




## Checks

- Part (ii) predicts $f(-1) - f(1) = 1$. Numerically $f(1) = 1/(\mathrm{e} - 1) \approx 0.58198$ and $f(-1) = \mathrm{e}/(\mathrm{e} - 1) \approx 1.58198$, which differ by exactly 1.
- The derivative formula evaluated at $t = \pm 10^{-4}$ gives $-0.49998$ and $-0.50002$, consistent with the limit $-\frac{1}{2}$.
- The coth form gives $g(t) \approx 1 + \frac{1}{12}t^2$ for small $t$, so $f(t) \approx 1 - \frac{1}{2}t + \frac{1}{12}t^2$, matching both limits in part (i).

## Common slips

- Using L'Hôpital once on $f'(t)$ and stopping at another $0/0$. Expanding both numerator and denominator to order $t^2$ avoids this.
- Proving only $\mathrm{e}^t(1 - t) \leq 1$ and then claiming $f'(t) \neq 0$. You need the inequality to be *strict* for $t \neq 0$, which comes from $t = 0$ being the only stationary point of $h$.
- Drawing the sketch through $(0, 1)$ as a solid point: $f$ is not defined there.

## Beyond STEP

The expansion $\dfrac{t}{\mathrm{e}^t - 1} = \sum_{n \geq 0} B_n \dfrac{t^n}{n!}$ defines the Bernoulli numbers $B_n$. Part (i) shows $B_0 = 1$ and $B_1 = -\frac{1}{2}$, and part (ii) shows that $B_n = 0$ for every odd $n \geq 3$. These numbers turn up in the Euler–Maclaurin summation formula, in sums of powers $1^k + 2^k + \cdots + n^k$, and in Euler's formula for $\sum 1/n^{2k}$. Removing a "hole" like the one at $t = 0$ is the idea of a removable singularity; for a careful treatment of limits and continuity, read Abbott, *Understanding Analysis*.

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